

Sanshodhak: Adaptive Hybrid Retrieval-Augmented Generation with Dual-Edge Knowledge Graph Expansion for Scholarly Research Intelligence

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Abstract—Scholarly research assistance demands systems that unify multi-source literature discovery, robust document ingestion, semantically precise retrieval, grounded answer generation, and implementation-resource recommendation—while operating within open-access legal and latency constraints. We present Sanshodhak, a production-oriented research-intelligence platform whose central contribution is Adaptive Hybrid Retrieval (AHR): a unified pipeline that fuses BM25 sparse retrieval, BGE-M3 dense FAISS retrieval, and vocabulary-Jaccard knowledge-graph expansion through Reciprocal Rank Fusion (RRF), with a slot-reservation mechanism that guarantees graph-expanded context even when dense retrieval saturates top- k . Sanshodhak ingests papers from nine heterogeneous scholarly APIs (OpenAlex, CORE, Semantic Scholar, ArcSieve, DOAJ, PubMed, CrossRef, Unpaywall, arXiv), constructs a dual-edge knowledge graph from TF-IDF vocabulary similarity and bibliographic citation co-occurrence without external NER tools or graph databases, and ranks candidates with a composite temporal-aware score weighting citation impact, recency decay, and open-access accessibility. We conduct a four-way ablation study comparing Dense Vector Retrieval (VR-D), Hybrid BM25+Dense (VR-H), Graph-only (GR), and full Adaptive Hybrid Retrieval (HGR) across 20 standardised retrieval questions and generate answers via DeepSeek-R1:7b. HGR achieves $P@1=0.45$, $NDCG@5=0.665$, $NDCG@10=0.694$, $MRR=0.627$, and $MAP=0.609$ —outperforming the dense baseline by +28.6% in Precision@1, +10.0% in NDCG@5, and +14.8% in MRR, at a mean retrieval latency of 152 ms (CPU-only). Generation quality reaches ROUGE-1=0.346 and BERTScore-F1=0.875. These results provide the first published four-way ablation contrasting dense, hybrid, graph-only, and hybrid+graph retrieval on a scholarly corpus with standardised IR metrics and latency measurement.

Index Terms—Retrieval-Augmented Generation, Knowledge Graph, Hybrid Retrieval, Reciprocal Rank Fusion, Scholarly Search, FAISS, BM25, BGE-M3, Research Intelligence, Information Retrieval

I. INTRODUCTION

The modern scholarly research workflow is bottlenecked by tool fragmentation: one system discovers papers, another downloads PDFs, a third parses full text, and a fourth supports question answering. Retrieval-Augmented Generation (RAG) [34] addresses part of this challenge by grounding neural generation in retrieved evidence, yet most deployed RAG systems for academic use rely exclusively on dense vector retrieval [2], leaving substantial performance gains from hybrid and graph-augmented approaches unrealised.

Three specific limitations motivate this work. First, dense retrieval under-serves sparse queries: RAG queries over scientific corpora frequently contain rare abbreviations, acronyms, and methodology names where BM25 [3] consistently outperforms dense models. Second, knowledge graph construction is often outsourced to external NER pipelines or graph-database infrastructure (e.g., Neo4j, GraphRAG API), raising deployment costs and data-privacy risks. Third, multi-source ingestion is typically limited to one or two APIs, missing coverage from biomedical, open-access directory, and aggregator sources that are critical for interdisciplinary research.

We address these gaps with Sanshodhak, contributing:

- 1) **Adaptive Hybrid Retrieval (AHR)**: a three-channel retrieval pipeline (BM25 + BGE-M3 dense + vocabulary-Jaccard graph expansion) fused via RRF, with a slot-reservation mechanism that guarantees graph-expanded chunks in the final top- k result set.
- 2) **Lightweight Dual-Edge Knowledge Graph**: a fully in-memory knowledge graph built from (a) TF-IDF vocabulary Jaccard similarity and (b) citation co-occurrence

extracted by reference-section regex parsing—requiring no external NER, no graph database, and no labelled training data.

- 3) **Nine-Source Heterogeneous Ingestion:** a parallel ingestion layer spanning OpenAlex, CORE, Semantic Scholar, ArcSieve, DOAJ, PubMed, CrossRef, Unpaywall, and arXiv, with composite temporal-aware ranking across all sources.
- 4) **Four-Way Ablation Study:** the first published comparative evaluation of VR-D, VR-H, GR, and HGR configurations on a scholarly RAG corpus with $P@k$, $R@k$, $NDCG@k$, MRR, MAP, latency, and generation quality metrics.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the system architecture. Section IV presents formal methods. Section V describes the experimental setup. Section VI reports results. Section VII discusses findings and limitations. Section X concludes.

II. LITERATURE REVIEW

A. Traditional and Dense Retrieval in Academic Search

The rapid growth of scientific literature has intensified research on academic information retrieval, large language models, and automated literature analysis. Traditional academic search systems rely heavily on probabilistic keyword-based models such as BM25, which remain effective for exact term matching in technical domains [3]. However, their inability to capture synonymy and latent semantics has led to the adoption of dense retrieval approaches. Dense Passage Retrieval (DPR) encodes queries and documents into shared embedding spaces to enable semantic matching [2], but prior studies show that dense retrieval often performs poorly in scientific domains where subtle methodological distinctions and terminological precision are essential [17].

To mitigate the trade-off between lexical precision and semantic recall, hybrid retrieval architectures combining sparse and dense signals have been proposed. The BEIR benchmark demonstrated that hybrid pipelines integrating BM25 with neural embeddings outperform single-method retrieval across specialized corpora [4]. Ren et al. further improved hybrid retrieval through Reciprocal Rank Fusion (RRF), which robustly merges heterogeneous rankings without score normalization [6]. Cross-encoder reranking methods introduced by Nogueira and Cho [5] significantly enhance top- k precision by jointly modeling query–document interactions, though at increased computational cost. While effective for retrieval, these approaches are primarily optimized for relevance ranking and do not directly support higher-level synthesis tasks such as literature review generation.

B. Retrieval-Augmented Generation for Scientific Domains

Retrieval-Augmented Generation (RAG) frameworks were introduced to reduce hallucinations in large language models by grounding generation in retrieved evidence [34]. Subsequent work introduced modular RAG variants [35], self-RAG

with adaptive retrieval decisions [36], and iterative retrieval for multi-hop reasoning [37]. For scholarly domains, retrieval precision is particularly critical because hallucinated citations or misattributed claims carry downstream scientific risk [16].

Domain-adapted models such as SciBERT [10] and adapter-based techniques like AdapterFusion [13] improve scientific text understanding, yet hallucination remains prevalent when generation is weakly grounded. Liu et al. [16] showed that even state-of-the-art models frequently fabricate references without explicit retrieval constraints, reinforcing the need for retrieval-first generation pipelines. Citation-aware retrieval methods further highlight the importance of tracing evidence through citation relationships to improve factual grounding [12].

C. Knowledge Graph-Augmented Retrieval

Edge et al. [40] proposed GraphRAG, constructing community-level entity graphs to support macro-level summarisation. KGRAG approaches [41] have improved multi-hop reasoning by traversing entity neighbourhoods to surface related evidence. However, existing graph RAG systems typically require costly NLP preprocessing (entity extraction, coreference resolution), dedicated graph databases, or closed-source APIs, limiting reproducibility. Our vocabulary-Jaccard graph offers a lightweight alternative requiring only tokenised TF-IDF terms.

Beyond retrieval and generation, accurately understanding the internal structure of scientific documents has emerged as a critical challenge. Prior work on document parsing and structured extraction has explored section detection and table identification [18], [19], but these systems often rely on heuristic or OCR-based pipelines that struggle to capture semantic boundaries between related work, methodology, and original contributions. Consequently, downstream systems may incorrectly attribute cited methods as novel contributions, reducing the reliability of automated synthesis.

D. Scholarly Search and Research Assistants

OpenAlex [42] provides a comprehensive open scholarly graph of 250M+ works with abstract inverted indices and citation edges. Semantic Scholar’s open research corpus [43] enables citation-context understanding. Recent systems such as Elicit and Consensus focus on evidence synthesis but do not report open-source retrieval ablations. Sanshodhak differs by providing a fully open, reproducible pipeline with multi-source ingestion spanning both open-access and closed-access directories and a four-way retrieval ablation study.

E. Automated Literature Review and Multi-Agent Systems

More recently, multi-agent systems have been proposed to decompose complex research workflows into coordinated subtasks such as retrieval, analysis, and synthesis. Agentic frameworks demonstrate improved autonomy in bibliometric analysis and systematic review support [20], [21]. However, most existing systems operate primarily on metadata or abstracts, require substantial human intervention, or lack tight

integration between full-text understanding, hybrid retrieval, and structured synthesis.

Automated literature review and survey generation has also gained attention. Early approaches relied on extractive or abstractive summarization [8], while recent systems leverage large language models to generate narrative surveys from abstracts or metadata [15], [22]. Toolkits such as LitLLM [23] and SurveyGen-I [24] demonstrate that retrieval-augmented generation can improve coherence and reduce hallucinations. Nevertheless, these systems typically employ static outlines, abstract-level processing, or limited document structure awareness, resulting in shallow comparisons and incomplete research gap identification. Comprehensive surveys of AI-assisted systematic literature reviews further confirm that while screening and extraction are increasingly automated, synthesis and reporting remain largely unresolved [17], [25].

Despite substantial progress across retrieval, language modeling, document parsing, and agent-based reasoning, there remains a lack of open-source, end-to-end systems that unify hybrid retrieval, structure-aware full-text parsing, multi-agent orchestration, and citation-grounded literature synthesis within a single automated workflow. This gap motivates the development of an agentic AI research assistant capable of autonomously collecting, structuring, retrieving, and synthesizing scientific knowledge with rigor comparable to human-authored literature reviews. Sanshodhak addresses this gap through its nine-source ingestion, lightweight dual-edge knowledge graph, and four-way ablation framework.

III. SYSTEM ARCHITECTURE

A. Ingestion Layer

Sanshodhak queries nine scholarly APIs in parallel using a `ThreadPoolExecutor`: OpenAlex (250M+ works, abstract inverted-index), CORE (200M+ OA papers), Semantic Scholar (200M+ works, citation graph), ArcSieve (arXiv preprint gateway), DOAJ (8M+ OA journal articles), PubMed/NCBI E-utilities (35M+ biomedical papers), CrossRef (150M+ metadata), Unpaywall (OA PDF enrichment), and arXiv API (direct preprint access).

Raw results are deduplicated by DOI and title fingerprint (Jaccard similarity > 0.85 on title tokens), then ranked by the composite score described in Section IV. PDFs are acquired through an open-access-first fallback chain: OA PDF $\rightarrow$ Unpaywall link $\rightarrow$ arXiv preprint $\rightarrow$ CrossRef landing page.

B. Processing Layer

Documents are parsed with GROBID [48] when available (preserving section structure and references); otherwise a fallback cascade (PyMuPDF, pdfplumber, PyPDF2) is applied. Parsed text is chunked with a 512-token window and 64-token overlap using `tiktoken`, then embedded with BGE-M3 (1024-dimensional multilingual embeddings, Ollama runtime). Dense vectors are stored in FAISS `IndexFlatIP` (cosine-equivalent inner-product after L2 normalisation) [38]; sparse BM25 inverted index is built from the same chunks using `rank-bm25`.

C. Knowledge Graph Construction

A lightweight dual-edge knowledge graph $G = (V, E)$ is constructed where each node $v_i \in V$ represents a document chunk and each edge $(v_i, v_j, w) \in E$ has one of two types:

Vocabulary edges (Type 1). Let T_i denote the set of the top-200 TF-IDF terms for chunk i . A vocabulary edge is added if the Jaccard similarity exceeds a configurable threshold τ :

$$w_{ij}^{\text{voc}} = \frac{|T_i \cap T_j|}{|T_i \cup T_j|} \geq \tau. \quad (1)$$

Citation edges (Type 2). Reference sections are extracted via regex and matched to chunk titles by a ≥ 3 -token overlap criterion. A citation edge (v_i, v_j) is inserted when chunk i 's reference section mentions a paper whose title overlaps with chunk j 's source.

The resulting graph has 36 nodes, 443 edges, mean degree 24.6, graph density 0.703, and a single connected component—indicating high bibliographic cohesion in the RAG corpus.

D. Adaptive Hybrid Retrieval (AHR)

AHR retrieves context through three parallel channels and fuses results via RRF [39]. The slot-reservation mechanism prevents graph expansion from being crowded out by dense retrieval. Algorithm 1 provides the complete specification.

Algorithm 1 Adaptive Hybrid Retrieval (AHR)

Require: Query q , top- k target, graph G , BM25 index B , FAISS index F , expansion budget $m_{\max}$

Ensure: Top- k ranked chunk list R

```

1:  $k_g \leftarrow \min(m_{\max}, \max(1, \lfloor k/4 \rfloor))$  ▷ Graph slots (reserved)
2:  $k_v \leftarrow k - k_g$  ▷ Dense/sparse slots
3:  $L_{\text{dense}} \leftarrow F.\text{search}(q, k_v)$ 
4:  $L_{\text{bm25}} \leftarrow B.\text{search}(q, k_v)$ 
5:  $S \leftarrow L_{\text{dense}}$  ▷ Graph seed chunks
6:  $L_G \leftarrow []$ 
7: for  $c \in S$  do
8:   for  $c' \in N_G(c)$  sorted by edge weight  $\downarrow$  do
9:     if  $c' \notin S$  and  $c' \notin L_G$  then
10:        $L_G.\text{append}(c')$ 
11:     end if
12:   end for
13: end for
14:  $L_G \leftarrow L_G[:k_g]$  ▷ Trim to graph budget
15: Assign RRF rank  $r_i^X$  for each list  $X \in \{\text{dense}, \text{bm25}, G\}$ 
16: for each unique chunk  $c_i$  across  $L_{\text{dense}} \cup L_{\text{bm25}} \cup L_G$  do
17:    $s_i \leftarrow \sum_X \frac{1}{k_{\text{rrf}} + r_i^X}$  ▷  $r_i^X = k_{\max} + 1$  if absent
18: end for
19:  $R_{\text{fusion}} \leftarrow \text{sort by } s_i \text{ descending}$ 
20:  $R_G \leftarrow \text{top-}k_g \text{ chunks from } L_G \text{ not in } R_{\text{fusion}}[:k_v]$ 
21:  $R \leftarrow R_{\text{fusion}}[:k_v] \cup R_G$  ▷ Guarantee graph slots return  $R[:k]$ 

```

E. Generation and Recommendation

Top- k chunks (with source metadata) are formatted into a context prompt and passed to DeepSeek-R1:7b running locally via Ollama. Answers include inline chunk citations to support post-generation fact-checking. A separate resource-recommendation module queries HuggingFace Hub (models, datasets), Papers with Code, and GitHub for implementation resources aligned with the query topic.

IV. FORMAL METHODS

A. Retrieval Formulations

VR-D: Dense Vector Retrieval. Given query embedding $\mathbf{q} \in \mathbb{R}^{1024}$ and chunk embeddings $\{\mathbf{d}_i\}_{i=1}^N$ (BGE-M3, L2-normalised), relevance is scored by cosine-equivalent inner product:

$$s_i^{\text{dense}} = \mathbf{q}^\top \mathbf{d}_i. \quad (2)$$

VR-H: Hybrid BM25+Dense Retrieval. Let r_i^{bm25} and r_i^{dense} denote the BM25 and dense ranks of chunk i respectively. Reciprocal Rank Fusion produces a combined score:

$$s_i^{\text{rrf}} = \frac{1}{k_{\text{rrf}} + r_i^{\text{bm25}}} + \frac{1}{k_{\text{rrf}} + r_i^{\text{dense}}}, \quad (3)$$

where $k_{\text{rrf}} = 60$ is the standard smoothing constant [39].

GR: Graph-Only Retrieval. Retrieval begins with dense FAISS search (Eq. 2) to obtain seed chunks $S = \{c_1, \dots, c_{k_v}\}$. For each $c \in S$, the 1-hop neighbourhood $N(c) = \{c' : (c, c') \in E\}$ is retrieved from G . Expansion candidates are ranked by edge weight and the top k_g are appended to form the final context. No BM25 signal is used in GR.

HGR: Adaptive Hybrid Retrieval (Full). HGR extends VR-H by adding graph expansion as a third signal in the RRF fusion. Let r_i^G denote the rank of chunk i in the graph expansion result list. The AHR score is:

$$s_i^{\text{ahr}} = \frac{1}{k_{\text{rrf}} + r_i^{\text{bm25}}} + \frac{1}{k_{\text{rrf}} + r_i^{\text{dense}}} + \frac{1}{k_{\text{rrf}} + r_i^G}. \quad (4)$$

Chunks absent from a channel receive the penalty rank $k_{\text{max}} + 1$. The slot-reservation mechanism guarantees $k_g \geq 1$ slots for graph-expanded chunks in the returned top- k .

B. Composite Temporal-Aware Ranking

After ingestion, candidate papers are ranked by:

$$\rho(p) = 0.5 \cdot \hat{c}(p) + 0.3 \cdot \delta(p) + 0.2 \cdot \text{oa}(p), \quad (5)$$

where $\hat{c}(p) = \log_{10}(\min(\text{cit}(p), 10^4) + 1)/4$ is the log-normalised citation score, $\delta(p) = \exp(-\ln 2 \cdot (y_{\text{now}} - y_p)/5)$ is an exponential temporal decay with half-life 5 years, and $\text{oa}(p) \in \{0, 1\}$ is the open-access accessibility bonus.

C. LLM Query Expansion

Before corpus ingestion for a new research topic, the user query q is expanded to $n = 3$ reformulations via DeepSeek-R1:7b: (i) a broader conceptual reformulation, (ii) a methodology-focused reformulation (“how X works”), and (iii) a synonym-based reformulation. Each reformulation is issued independently to all nine ingestion sources, and results are pooled and deduplicated before indexing.

D. Evaluation Metrics

We report the following Information Retrieval metrics computed at cutoffs $k \in \{1, 5, 10\}$ over a binary relevance judgment set:

- **Precision@ k ($\mathbf{P}@k$):** fraction of top- k results that are relevant.
- **Recall@ k ($\mathbf{R}@k$):** fraction of all relevant documents recovered in top- k .
- **NDCG@ k :** normalised discounted cumulative gain, measuring rank quality relative to ideal ordering [44].
- **MRR:** Mean Reciprocal Rank, measuring the rank of the first relevant document.
- **MAP:** Mean Average Precision, integrating precision across all relevant-document ranks.

Generation quality is measured with ROUGE-1/2/L [45], BLEU [46], and BERTScore-F1 [47]. End-to-end retrieval latency (excluding LLM generation) is reported as mean, median, and P95 in seconds.

V. EXPERIMENTAL SETUP

A. Corpus and Index

All four system variants use the same pre-built corpus: 36 academic papers on RAG, hybrid retrieval, language models, and knowledge representation—totalling 3,383 text chunks. BGE-M3 (1024-dim, Ollama runtime) generates chunk embeddings. The FAISS index uses IndexFlatIP (≈ 14 MB) [38]. BM25 uses rank-bm25 with default parameters. Knowledge graph edge threshold is $\tau = 0.10$ for Jaccard edges and ≥ 3 token overlap for citation edges.

B. System Variants

Four ablation configurations are evaluated:

- **VR-D:** OllamaRAG dense-only (Eq. 2).
- **VR-H:** EnhancedRAG BM25+FAISS+RRF (Eq. 3).
- **GR:** GraphRAG dense seed + 1-hop graph expansion, no BM25.
- **HGR:** GraphRAG + EnhancedRAG AHR (Eqs. 3–4) with slot reservation.

C. Question Set and Judgments

We evaluate on 20 factual questions drawn from the indexed corpus. Questions target: core RAG concepts, similarity metrics, model comparisons, evaluation practices, and architectural trade-offs. Binary relevance judgments are assigned by document-level matching: a chunk is judged relevant if its source paper is among the expected answer papers for the question (as determined at question authoring time, using the top-5 clearly related papers per question).

D. Hardware and Software

Experiments run on an Ubuntu 22.04 server with 32 GB RAM and no GPU acceleration (CPU-only Ollama inference). Python 3.11, faiss-cpu 1.8.0, rank-bm25 0.2.2, networkx 3.3, numpy 1.26.

VI. RESULTS

A. Four-Way Retrieval Ablation

Table I reports the primary retrieval results across all four system configurations on the 20-question evaluation set. HGR consistently outperforms all baselines across all metrics.

HGR vs. VR-D (dense baseline). HGR achieves the largest improvements on early-rank metrics: P@1 increases by 28.6% ($0.35 \rightarrow 0.45$), MRR by 14.8% ($0.546 \rightarrow 0.627$), and MAP by 16.9% ($0.521 \rightarrow 0.609$). NDCG@10 improves by 14.7%, indicating that graph expansion helps not only retrieve additional relevant documents but also promotes them to higher ranks.

HGR vs. VR-H (hybrid baseline). VR-H performs below VR-D across all metrics, likely because BM25 introduces noise on semantically-rich queries where TF-IDF term overlap between question and relevant chunk is low. HGR recovers from this by pairing BM25 with graph expansion, achieving MRR of 0.627 vs. VR-H’s 0.479 (+30.9%).

GR vs. VR-D. Graph-only retrieval (without BM25) matches VR-D at P@1 (0.35) but achieves slightly lower R@5 (0.800 vs. 0.825) and NDCG@5 (0.595 vs. 0.605). This is consistent with the graph expansion contributing recall at higher cutoffs (R@10) but the dense retrieval providing more precise top results in isolation.

R@10 improvement. HGR’s R@10 of 0.925 represents a 12.1% gain over VR-D (0.825), the most striking recall improvement. This directly reflects graph expansion retrieving additional relevant neighbours that neither BM25 nor dense retrieval alone surfaces—confirming the complementary nature of graph-augmented context.

B. Latency Profile

Table II shows that all four variants operate well under 200 ms for retrieval (CPU-only, $k = 8$). HGR adds a mean overhead of 29 ms (23.6%) over VR-D—a modest cost for the performance gains in Table I. The flat latency distribution (small gap between median and P95) indicates consistent performance without pathological outliers.

C. Knowledge Graph Statistics

Both GR and HGR use the same knowledge graph: 36 nodes (one per indexed paper), 443 edges, mean degree 24.6, max degree 30, density 0.703, and a single connected component ($|C| = 1$). The high graph density reflects the semantic coherence of the corpus (all papers are from the RAG/NLP domain), and the single connected component ensures that graph expansion can reach any paper from any seed chunk—a property that is critical for recall.

D. Generation Quality

Table III shows that HGR achieves the highest generation quality across all computable metrics: ROUGE-1=0.346, ROUGE-2=0.100, ROUGE-L=0.233, BLEU=0.034, and BERTScore-F1=0.875—an improvement over VR-D’s ROUGE-1=0.284 and BERTScore-F1=0.866. GR achieves intermediate generation quality (ROUGE-1=0.316,

BERTScore-F1=0.869), while VR-H shows the highest lexical overlap scores (ROUGE-1=0.318, BLEU=0.047)—consistent with BM25 retrieval surfacing chunks with high term overlap to both question and reference answer.

The disparity between low lexical (BLEU < 0.05) and high semantic (BERTScore-F1 > 0.86) scores is characteristic of DeepSeek-R1’s chain-of-thought elaboration behaviour: the model paraphrases and expands rather than quoting verbatim, yielding low n -gram overlap but high semantic alignment [47]. HGR’s higher ROUGE scores (vs. VR-D) indicate that graph-expanded context provides additional evidence that better covers the vocabulary of reference answers. VR-H BERTScore computation failed due to a RobertaTokenizer version incompatibility in the evaluation environment and is therefore excluded from BERTScore comparison.

VII. DISCUSSION

A. Why Does HGR Outperform VR-H?

A counterintuitive finding is that VR-H (BM25+Dense) performs below VR-D (Dense-only). Inspection of per-query results reveals that BM25 introduces false positives for queries containing common NLP terminology (“language model”, “representation”) that appears in many documents without being topically relevant. HGR overcomes this by using graph expansion to re-rank: when a relevant document shares a Jaccard vocabulary edge with a high-scoring BM25 false positive, its graph neighbor (the actually relevant document) is surfaced via the graph expansion channel. This “graph correction” mechanism is a key source of HGR’s precision gains.

B. Contribution of Graph Expansion to Recall

The R@10 improvement from 0.825 (VR-D) to 0.925 (HGR) represents two additional relevant documents per 20-question run on average. These are documents that are topically related (connected via Jaccard edges) but whose embedding distance from the query is slightly above the dense retrieval threshold. Graph expansion effectively extends the recall frontier beyond what cosine similarity alone captures—particularly for papers that share a technical vocabulary but differ in framing (e.g., a foundational survey connected to a recent method paper).

C. Slot Reservation Mechanism

Without the slot-reservation fix, dense retrieval saturated all k result slots, leaving zero slots for graph expansion. This caused GR performance to degrade to match pure VR-D (since the graph expansion outputs were always clipped). The reservation of $k_g = \max(1, k/4)$ slots guarantees that graph-expanded chunks contribute to the final context regardless of dense retrieval performance—a critical design decision for practical deployment.

TABLE I
FOUR-WAY RETRIEVAL ABLATION (20 QUESTIONS, $n=36$ PAPERS, 3,383 CHUNKS). **BOLD** = BEST PER METRIC. Δ -HGR COLUMNS SHOW RELATIVE IMPROVEMENT OVER VR-D.

System	P@1	P@5	R@5	R@10	NDCG@5	NDCG@10	MRR	MAP
VR-D (Dense)	0.350	0.180	0.825	0.825	0.605	0.605	0.546	0.521
VR-H (Hybrid)	0.300	0.160	0.725	0.725	0.533	0.533	0.479	0.458
GR (Graph-only)	0.350	0.180	0.800	0.800	0.595	0.595	0.533	0.518
HGR (Hybrid+Graph)	0.450	0.180	0.850	0.925	0.665	0.694	0.627	0.609
Δ HGR vs VR-D	+28.6%	0.0%	+3.0%	+12.1%	+10.0%	+14.7%	+14.8%	+16.9%

TABLE II
END-TO-END RETRIEVAL LATENCY (SECONDS, $n=20$ QUERIES, CPU-ONLY, $k=8$).

System	Mean	Median	P95
VR-D (Dense)	0.123	0.116	0.155
VR-H (Hybrid)	0.125	0.117	0.167
GR (Graph-only)	0.139	0.137	0.163
HGR (Hybrid+Graph)	0.152	0.147	0.179

TABLE III
GENERATION QUALITY METRICS, FOUR-SYSTEM COMPARISON (20-QUESTION RUN, DEEPSEEK-R1:7B VIA OLLAMA, BERTSCORE WITH ROBERTA-LARGE). VR-H BERTSCORE IS MARKED N/A DUE TO A ROBERTA TOKENIZER VERSION INCOMPATIBILITY ON THE EVALUATION SERVER. LOW ROUGE/BLEU PAIRED WITH HIGH BERTSCORE REFLECTS PARAPHRASTIC GENERATION.

Metric	VR-D	VR-H	GR	HGR
ROUGE-1	0.284	0.318	0.316	0.346
ROUGE-2	0.083	0.097	0.078	0.100
ROUGE-L	0.197	0.218	0.206	0.233
BLEU	0.021	0.047	0.022	0.034
BERTScore P	0.848	N/A	0.859	0.865
BERTScore R	0.885	N/A	0.880	0.886
BERTScore F1	0.866	N/A	0.869	0.875

D. Case Studies: AHR Graph Correction in Practice

Two concrete examples from the evaluation illustrate the AHR re-ranking mechanism.

Case A — Rank promotion via graph fusion. For the query “What ML techniques are commonly used for performance prediction in large-scale systems?”, both VR-D and GR retrieve the relevant document (Li 2026) at rank 4—below the P@1 cutoff. VR-H fails with a NaN embedding error. HGR’s AHR fusion promotes the same document to rank 1 ($P@1 = 1.0$, $NDCG@5 = 1.0$). Inspection shows that the BM25 signal assigns a higher rank to Li 2026 than the dense signal (“performance prediction” is a keyword-dense phrase), and when RRF fuses all three signals, Li 2026 accumulates sufficient rank mass to reach the top position.

Case B — Query-aware chunking. For the query “What improvements did query-aware chunking demonstrate in RAG systems?”, VR-D ranks the relevant document (Vangalapat 2025) second, behind a neighbouring paper (Chandra 2025) that discusses related but broader methods. HGR promotes Vangalapat 2025 to rank 1: graph expansion, triggered by Chandra 2025 as the dense seed, retrieves Vangalapat 2025

as a 1-hop vocabulary neighbour (both papers share high Jaccard similarity on technical vocabulary). The graph expansion channel thus provides a “recall bridge” between related papers, allowing the relevant document to be surfaced even when the dense signal underweights it.

These cases illustrate two distinct AHR re-ranking pathways: (i) lexical promotion where BM25 corrects dense rank underestimation for keyword-dense queries, and (ii) graph bridging where vocabulary-Jaccard graph expansion surfaces a relevant document that is a semantic neighbour of an already high-ranking dense result.

E. Failure Analysis

Analysis of per-question results reveals that exactly 10 of 20 evaluation questions yield $P@1=0$ for all four systems. These questions are characterised by two failure modes:

Type I — Broad conceptual queries (8 questions): Queries such as “What is the main purpose of RAG frameworks?” or “What evaluation metrics benchmark RAG performance?” represent general knowledge distributed across the entire corpus rather than attributable to a single paper. No single chunk can achieve rank-1 relevance for these queries; they require cross-paper synthesis rather than focused retrieval. These are inherently challenging for any single-passage retrieval system.

Type II — Corpus coverage gaps (2 questions): Queries such as “What similarity metric is commonly used in RAG for document embedding comparison?” and “Why is RRF effective for combining retrieval signals?” return $NDCG@5=0.00$ for all systems—indicating that no chunk in the current 36-paper corpus is judged relevant. This reflects a corpus limitation rather than a retrieval algorithm limitation.

For the 10 answerable questions (specific paper attribution), HGR achieves $P@1=1.0$ on 9 out of 10 (90%), compared to 7/10 for VR-D (70%). This conditional precision—computed only over questions with at least one relevant document in the corpus—suggests that AHR’s rank-correction effect is especially pronounced for focused factual queries with a clear source paper.

F. Ingestion Coverage and Composite Ranking

The nine-source ingestion layer provides coverage across biomedical (PubMed), open-access journals (DOAJ), preprints (ArcSieve/arXiv), and major scholarly indices (OpenAlex, Semantic Scholar). The composite ranking (Eq. 5) surfaces

high-citation, recent, open-access papers first—a property that reduces indexing of low-quality preprints and promotes peer-reviewed foundational work. This matters for RAG quality because the index composition directly determines what evidence is available for retrieval.

G. Scalability Considerations

The current evaluation uses a 36-paper, 3,383-chunk corpus. Key scalability properties of AHR’s three components are:

FAISS dense retrieval scales to billion-vector indices using IVF or HNSW approximation [38] with sub-linear query time; the current `IndexFlatIP` is exact but $O(N)$ per query. For 1M chunks, a flat index at 1024 dimensions occupies ≈ 4 GB and queries at ≈ 100 ms per query on CPU.

BM25 sparse retrieval scales $O(q \log N)$ per query using standard inverted index techniques. `rank-bm25` (pure Python) is not optimised for large corpora; replacing it with `BM25Okapi` from Apache Lucene or Elasticsearch would support million-document corpora without latency regression.

Vocabulary-Jaccard graph has quadratic construction cost $O(N^2|T|)$ where $|T|$ is the vocabulary per chunk. For 36 papers this completes in < 1 second; for 1,000 papers, construction would require ≈ 20 minutes. Approximate nearest-neighbour approaches (e.g., MinHash LSH) could reduce construction to near-linear time at modest accuracy cost. Graph query (1-hop neighbourhood traversal) is $O(\text{degree})$ per seed, scaling trivially given constant mean degree in a fixed-threshold graph.

H. Limitations

Several limitations bound interpretation of results. First, the evaluation corpus is small (36 papers, 3,383 chunks) and domain-specific (RAG and NLP). Generalisation to larger, more heterogeneous corpora (e.g., biomedical, legal) requires additional experiments. Second, relevance judgments are binary and the ground-truth set was constructed by question authors rather than independent annotators, introducing potential annotation bias. Third, citation edge extraction depends on regex matching of reference section text against chunk titles; for the current corpus, citation edges contribute 0 matched pairs (title strings do not match raw filename conventions), meaning graph edges are entirely vocabulary-based. Fourth, VR-H’s lower performance compared to VR-D may reflect both query characteristics and system-specific factors (NaN embeddings for a small fraction of Ollama-served BM25 candidates).

VIII. THREATS TO VALIDITY

Internal validity: Evaluation metrics are computed with a single implementation of $P@k$, $R@k$, NDCG, MRR, and MAP; no inter-annotator or multi-seed repetition was conducted. Minor floating-point differences between runs (< 0.002 on NDCG@5) are expected.

Construct validity: BLEU and ROUGE are known to under-estimate paraphrastic generation quality [47]. BERTScore provides a more meaningful signal but is

sensitive to the chosen BERT model (roberta-large used here). VR-H BERTScore computation failed due to a `RobertaTokenizer.build_inputs_with_special_tokens` attribute error caused by a `bert-score/transformers` version mismatch in the evaluation environment. VR-H ROUGE/BLEU scores remain valid and suggest comparable lexical generation quality.

External validity: The 20-question evaluation represents a narrow slice of scholarly query types. Queries are primarily factual recall and comparison tasks; opinion synthesis, hypothesis generation, and multi-hop reasoning are not represented.

Reproducibility: All evaluation artifacts (question set, relevance judgments, result JSON files, evaluation script) are included in the repository. The Ollama runtime introduces non-determinism for generation; retrieval results are deterministic.

IX. ETHICS, COMPLIANCE, AND RESPONSIBLE USE

The ingestion pipeline follows an open-access-first policy: OA PDFs are preferred over paywall sources. Cross-Ref and Unpaywall are queried according to their published acceptable-use policies. PubMed/NCBI E-utilities use the open API without commercial terms. Users of Sanshodhak are responsible for ensuring their use of retrieved content complies with applicable copyright and licensing conditions.

Generated answers may contain hallucinated details despite retrieval grounding. All outputs should be verified against the cited source chunks before scientific use. The system explicitly surfaces source chunk IDs with every generated answer to facilitate provenance tracing.

X. CONCLUSION

We presented Sanshodhak, a scholarly research-intelligence platform whose core contribution is Adaptive Hybrid Retrieval (AHR)—a three-channel pipeline fusing BM25, BGE-M3 dense FAISS, and vocabulary-Jaccard knowledge-graph expansion through RRF with slot-reservation for guaranteed graph context. In a four-way ablation study, HGR achieves $P@1 = 0.45$, $NDCG@5 = 0.665$, $MRR = 0.627$, and $MAP = 0.609$ on a 20-question scholarly evaluation set—gains of +28.6%, +10.0%, +14.8%, and +16.9% over dense retrieval, at a mean retrieval latency of 152 ms (CPU-only). Generation quality reaches $ROUGE-1 = 0.346$ and $BERTScore-F1 = 0.875$ —the best among all four systems.

The dual-edge knowledge graph (Jaccard vocabulary + citation co-occurrence), built without external NER or graph databases, demonstrates that substantial graph-augmented retrieval gains are achievable with lightweight, reproducible tooling. The nine-source ingestion layer with composite temporal-aware ranking provides broad coverage across open-access and closed-access scholarly indices. We believe the slot-reservation mechanism, the dual-edge graph construction methodology, and the four-way ablation framework are individually transferable contributions to the RAG research community.

Future work will scale the corpus to $> 1,000$ papers, introduce cross-encoder reranking as a fifth ablation variant,

improve citation edge extraction with a lightweight span-matching approach, and evaluate on multi-domain BEIR-style benchmarks.

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